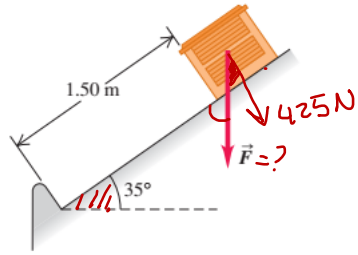
Q&A 01 Newton's Law

Monday, November 13, 2023 6:29 PM

An air-conditioning unit is fastened to a roof that slopes at an angle of 35° above the horizontal (Fig. 1.33). Its weight is a force on the air conditioner that is directed vertically downward. In order that the unit not crush the roof tiles, the component of the unit's weight perpendicular to the roof cannot exceed 425 N. (One newton, or 1 N, is the SI unit of force. It is equal to 0.2248 lb.) (a) What is the maximum allowed weight of the unit? (b) If the fasteners fail, the unit slides 1.50 m along the roof before it comes to a halt against a ledge. How much work does the weight force do on the unit during its slide if the unit has the weight calculated in part (a)? As we described in Section 1.10, the work done by a force $\vec{F}$ on an object that undergoes a displacement $\vec{s}$ is $W = \vec{F} \cdot \vec{s}$.

1.33 An air-conditioning unit on a slanted roof.



$F\cos 35 = 425$ $\quad F = 518.8N$

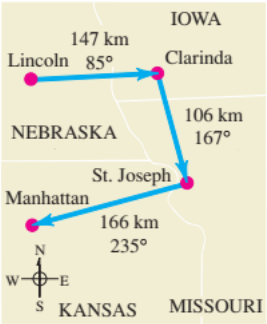
$W = \vec{F} \cdot \vec{s}$

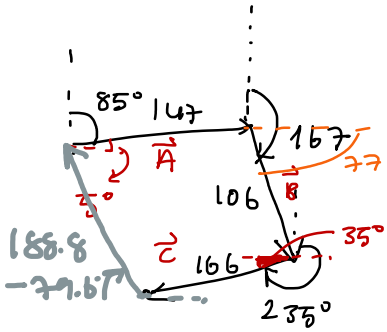
$= 518.8 \times 1.5 \times \cos 55$

$= 446.N$

1.74 ••• On a training flight, a student pilot flies from Lincoln, Nebraska, to Clarinda, Iowa, then to St. Joseph, Missouri, and then to Manhattan, Kansas (Fig. P1.74). The directions are shown relative to north: 0° is north, 90° is east, 180° is south, and 270° is west. Use the method of components to find (a) the distance she has to fly from Manhattan to get back to Lincoln, and (b) the direction (relative to north) she must fly to get there. Illustrate your solutions with a vector diagram.

Figure **P1.74**





$\vec{A} + \vec{B} + \vec{C} + \vec{D} = 0$

$\vec{D} = -(\vec{A} + \vec{B} + \vec{C}$

$\vec{A} = A_x\hat{i} + A_y\hat{j}$

$= A\cos 5\hat{i} + A\sin$

$\vec{A} = 146.4\hat{i} + 12.$

$\vec{B} = 106\cos 77\hat{i} + 106\sin 77(-\hat{j})$

$= 23.8\hat{i} - 103.3\hat{j}$

$\vec{C} = 166\cos 35(-\hat{i}) + 166\sin 35$

$= -136.0\hat{i} - 95.2\hat{j}$

$\vec{D} = -[(146.4 + 23.8 - 136)\hat{i} + (12.8 - 103.3 - 95.2)\hat{j}]$

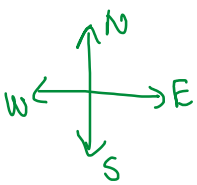
$\vec{D} = -34.2\hat{i} + 185.7\hat{j}$

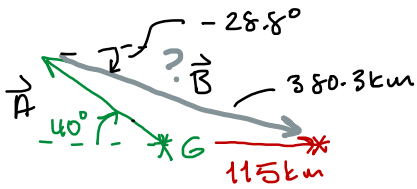
$|\vec{D}| = 188.8\,km$ $\quad \theta = \tan^{-1}\left(\frac{185.7}{-34.2}\right) = -79.6°$

1.78 ••• A ship leaves the island of Guam and sails 285 km at 40.0° north of west. In which direction must it now head and how far must it sail so that its resultant displacement will be 115 km directly east of Guam?

40° North of west

Batının 40° kuzeyi





$\vec{A} = 285\,co$

$= -218.$

$\vec{C} = 115\hat{i}$

$\vec{A} + \vec{B} = \vec{C}$

$\vec{B} = \vec{C} - \vec{A} = 115\hat{i} - (-218.3\hat{i}$

$\vec{B} = 333.3\hat{i} - 183.2\hat{j}$

$|\vec{B}| = 380.3\,km$ $\quad \theta = \tan^{-1}\left(\frac{-183.}{333.}\right.$

1.87 •• Vectors $\vec{A}$ and $\vec{B}$ have scalar product −6.00 and their vector product has magnitude +9.00. What is the angle between these two vectors?

$\vec{A} \cdot \vec{B} = -6 = AB\cos\theta$

$|\vec{A} \times \vec{B}| = 9 = |AB\sin\theta|$

$\frac{|\vec{A} \times \vec{B}|}{\vec{A} \cdot \vec{B}} = \frac{9}{-6} = \tan\theta$

$\theta = \tan^{-1}(-3/2) = -56.3°$

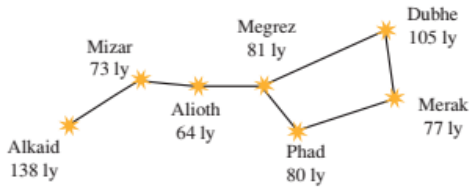
$\vec{A} \cdot \vec{B} < 0$

$\vec{B}$ 56.3

$\alpha = 123.7°$

1.101 ••• **Navigating in the Big Dipper.** All the stars of the Big Dipper (part of the constellation Ursa Major) may appear to be the same distance from the earth, but in fact they are very far from each other. Figure P1.101 shows the distances from the earth to each of these stars. The distances are given in light-years (ly), the distance that light travels in one year. One light-year equals 9.461×10^{15} m. (a) Alkaid and Merak are 25.6° apart in the earth's sky. In a diagram, show the relative positions of Alkaid, Merak, and our sun. Find the distance in light-years from Alkaid to Merak. (b) To an inhabitant of a planet orbiting Merak, how many degrees apart in the sky would Alkaid and our sun be?

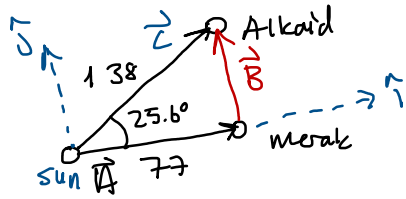
Figure **P1.101**



S = E

O

8 light min



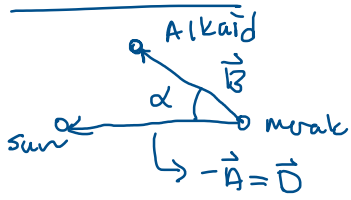
$\vec{A} + \vec{B} = \vec{C}$

$77\hat{i} + \vec{B} = 138 \cos 25.6$

$\vec{B} = (124.5 - 77)$

$\vec{B} = 47.5\hat{i} +$

$|\vec{B}| = 79.2$ ly ✓



$\hookrightarrow -\vec{A} = \vec{D}$

$\vec{D} \cdot \vec{B} = DB$

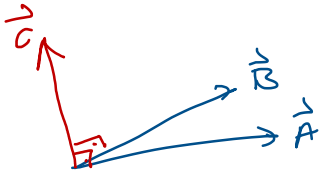
$-77\hat{i} \cdot (47.5\hat{i} -$

$\hookrightarrow \dfrac{-77\,(47.5)}{77\,(79.2)} = \cos\alpha \qquad \alpha = \cos^{-1}\left(-\dfrac{47.5}{79.2}\right) = 53.1°$

1.53 • Given two vectors $\vec{A} = -2.00\hat{i} + 3.00\hat{j} + 4.00\hat{k}$ and $\vec{B} = 3.00\hat{i} + 1.00\hat{j} - 3.00\hat{k}$, do the following. (a) Find the magnitude of each vector. (b) Write an expression for the vector difference $\vec{A} - \vec{B}$ using unit vectors. (c) Find the magnitude of the vector difference $\vec{A} - \vec{B}$. Is this the same as the magnitude of $\vec{B} - \vec{A}$? Explain.

$\vec{A}$ ve $\vec{B}$ dik olan birim vektor $\hat{C}$ which is $\hat{C} \perp \vec{A}$ and $\hat{C} \perp \vec{B}$.

$\vec{A} \times \vec{B} = \vec{C}$



$\dfrac{\vec{C}}{|\vec{C}|} = \hat{C}$ ✓

$\vec{A} \times \vec{B} = (-2\hat{i} + 3\hat{j} + 4\hat{k}) \times (3\hat{i} + \hat{j}$

$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$

$\hat{i} \times \hat{j} = +\hat{k}$

$\hat{j} \times \hat{k} = +\hat{i}$

$\hat{k} \times \hat{i} = +\hat{j}$

$= -2(\hat{i} \times \hat{j}) +$

$+ 9(\hat{j} \times \hat{i})$

$+ 12(\hat{k} \times \hat{i})$

$\vec{A} \times \vec{B} = (-2-9)\hat{k} + 6\hat{j} - 13\hat{i} = -13\hat{i} + 6\hat{j} - 11\hat{k} = \vec{C}$; $|\vec{C}| = \sqrt{13^2 +}$

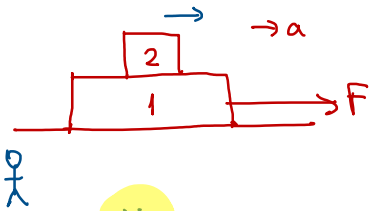
$\hat{C} = \dfrac{\vec{C}}{|\vec{C}|} = -\dfrac{13}{18.1}\hat{i} + \dfrac{6}{18.1}\hat{j} - \dfrac{11}{18.1}\hat{k}$

sınavda f_s, f_k ✓ sorulusunuz ; Ch3 ⇒ circular motion ✓

Daha dağıl { ama circular motion in Newton's law. hava sürtünmesi (air friction)

FBD, a ...? X

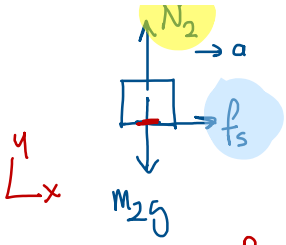
Application of Newton's Laws: (Ch 5)



1 & 2 are moving together with the same acceleration $\underline{a}$.

m_1 m_2 F μ_k μ_s g are given.

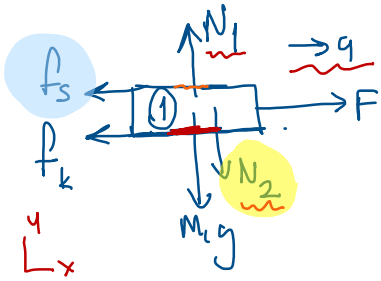
→ FBD of each object? → $a = ?$ $= \mu_s g$



2 & 1 surface is in contact.

f_s = static

→ What's the maximum of F value so that they move together save acceleration?



FBD 2 + FBD 1 =

$\Sigma F_y = m_2 \cancel{a_y}^{\,0}$

$N_2 = m_2 g$

$f_s = \mu_s N_2$

$\Sigma F_x = m_2 a_x$

$f_s = \mu_s m_2 g$

$f_s = m_2 a$

$\mu_s m_2 g = m_2 a$

$\boxed{a = \mu_s g}$

a_{max}

$\Sigma F_y = m_1 \cancel{a_y}^{\,0}$

$N_1 - N_2 - m_1 g = 0$

$N_1 = N_2 + m_1 g = (m_2 + m_1) g$

$\Sigma F_x = m_1 a_x = m_1 a$

$F - f_s - f_k = m_1 a$

$F - \mu_s N_2 - \mu_k N_1 = m_1 a$

$F - \mu_s m_2 g - \mu_k (m_1 + m_2) g = m_1 a$

$$\boxed{a = \frac{F - \mu_s m_2 g - \mu_k (m_1 + m_2) g}{m_1}}$$

$\Sigma F_x = m$

$F - f_k$

$F - \mu_k N$

$\Sigma F_y =$

$N_1 = (m$

$F - \mu_k (m$

$a = \frac{F}{m_1 +}$

yarum c

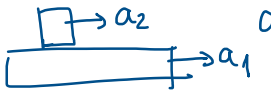
$$\mu_s g = \frac{F - \mu_s m_2 g - \mu_k (m_1 + m_2) g}{m_1}$$

$\mu_s m_1 g = F - \mu_s m_2 g - \mu_k (m_1 + m_2) g$

$F = \mu_s (m_1 + m_2) g + \mu_k (m_1 + m_2) g$

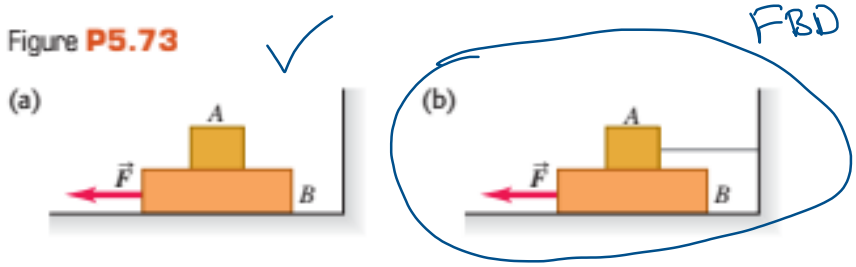
$F_{max} = (\mu_s + \mu_k)(m_1 + m_2) g$ → max force.

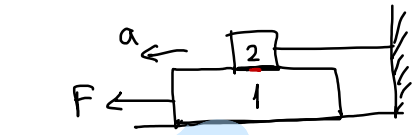
$\{ F_1 > F_{max}$



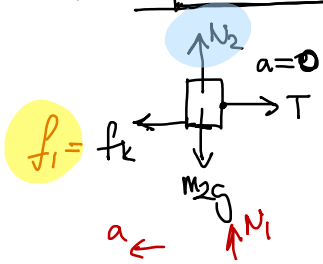
5.73 •• Block A in Fig. P5.73 weighs 2.40 N and block B weighs 3.60 N. The coefficient of kinetic friction between all surfaces is 0.300. Find the magnitude of the horizontal force $\vec{F}$ necessary to drag block B to the left at constant speed (a) if A rests on B and moves with it (Fig. P5.73a). (b) If A is held at rest (Fig. P5.73b).

Figure **P5.73**





FBD ? a? Tension on rope?



$\Sigma F_y = m_2 \cancel{a_y}^{\,0}$ $\quad N_2 = m_2 g$

$\Sigma F_x = m_2 \cancel{a_x}^{\,0}$ $\quad T = f_k = \mu_k N_2 = \mu_k m_2 g$

$\Sigma F_y = m_1 \cancel{a_y}^{\,0} \Rightarrow N_1 - N_2 - m_1 g = 0$

$$N_1 = (m_1+m_2)g$$

$$\Sigma F_x = m_1 a_x$$

$$F - f_1 - f_2 = m_1 a$$

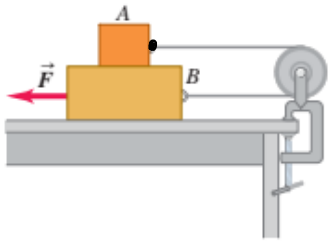
$$F - \mu_k N_1 - \mu_k N_2 = m_1 a$$

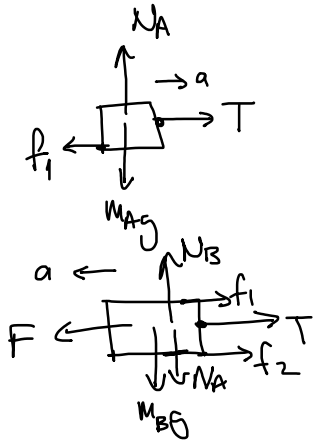
$$F - \mu_k (m_1+m_2)g - \mu_k m_2 g = m_1 a$$

$$a = \frac{F - \mu_k(m_1 + 2m_2)g}{m_1}$$

5.89 ••• Block A in Fig. P5.89 weighs 1.90 N, and block B weighs 4.20 N. The coefficient of kinetic friction between all surfaces is 0.30. Find the magnitude of the horizontal force $\vec{F}$ necessary to drag block B to the left at constant speed if A and B are connected by a light, flexible cord passing around a fixed, frictionless pulley.

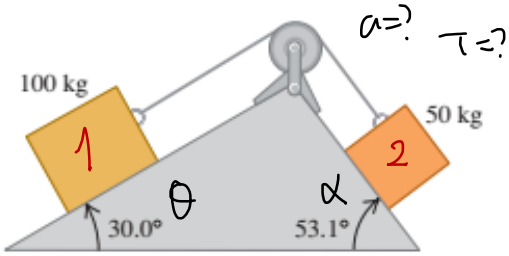
Figure **P5.89**



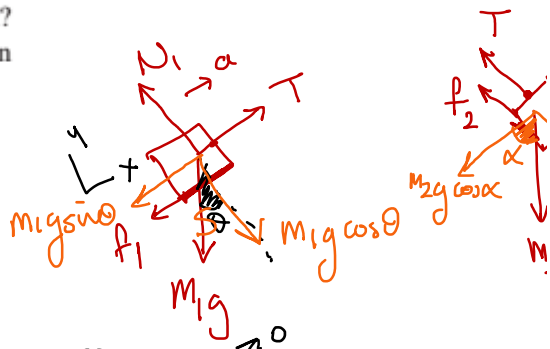


5.92 •• Two blocks connected by a cord passing over a small, frictionless pulley rest on frictionless planes (Fig. P5.92). (a) Which way will the system move when the blocks are released from rest? (b) What is the acceleration of the blocks? (c) What is the tension in the cord?

Figure **P5.92**



FBD



$$\Sigma F_y = m_1 \cancel{a_y}^{\,0}$$

$$N_1 = m_1 g\cos\theta$$

$$\Sigma F_x = m_1 a_x \qquad \mu N_1$$

$$T - m_1 g\sin\theta - f_1 = m_1 a$$

$$T - m_1 g\sin\theta - \mu m_1 g\cos\theta = m_1 a \quad (1)$$

$$\Sigma F_y = m_2 \cancel{a_y}^{\,0} \qquad N_2 = m_2 g\cos\theta$$

$$\Sigma F_x = m_2 a_x = m_2 a \quad ; f_2 = \mu N_2$$

$$m_2 g\sin\alpha - T - f_2 = m_2 a$$

$$m_2 g\sin\alpha - T - \mu m_2 g\cos\alpha = m_2 a \quad (2)$$

$$(1) + (2) \Rightarrow m_2 g\sin\alpha - m_1 g\sin\theta - \mu m_2 g\cos\alpha - \mu m_1 g\cos\theta = (m_1$$

$$(m_2\sin\alpha - m_1\sin\theta) - \mu(m_2\cos\alpha + m_1\cos\theta)$$

$u = \frac{\quad}{m_1 + m_2} g$; $T =$

5.97 • A block is placed against the vertical front of a cart as shown in Fig. P5.97. What acceleration must the cart have so that block *A* does not fall? The coefficient of static friction between the block and the cart is μ_s. How would an observer on the cart describe the behavior of the block?

$a_y = 0$

Figure **P5.97**

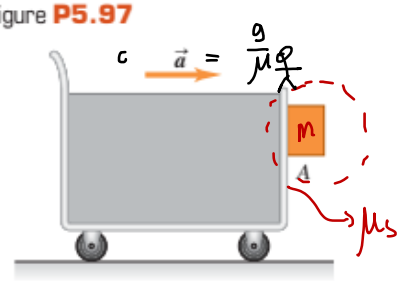
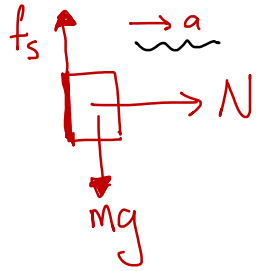


Figure **P5.98**

8.00

FBD



y
x

$\Sigma F_y = m$

$f_s - mg$:

$f_s = mc$

$\mu N =$

$N =$

$\rightarrow x$

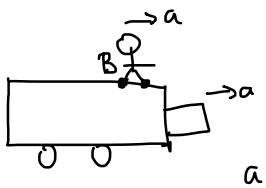
$\Sigma F_x = ma_x$

$N = ma$

$\mu < 1 \quad a > g$

$$\boxed{a = \frac{g}{\mu}}$$

$\Leftarrow \frac{mg}{\mu} = ma$

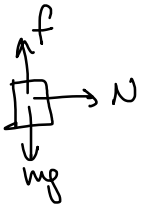
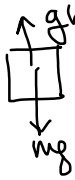


A

↳ acc. to A

accordy to B mass does not fall down?

ghost force



ghost force = friction =